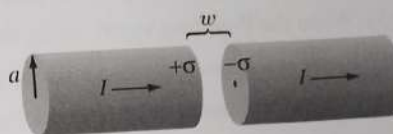


Final examination

(The meanings of the variables and constants are the same as those explained in the lectures.)

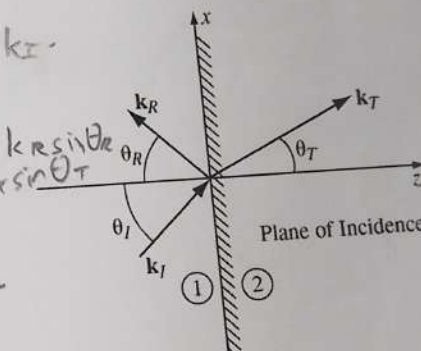
1. Write Maxwell's equations in matter (four equations including E , D , B , H , ρ_f and J_f).
2. Answer the following problems about electromagnetic fields in vacuum.
 - (1) Define the total energy stored in per unit volume using electric and magnetic fields and related constants.
 - (2) Define the Poynting vector using electric and magnetic fields and related constants.

3. Consider the charging capacitor in the figure.



- (1) Find the electric and magnetic fields in the gap, as functions of the distance s from the axis and the time t , assuming the charge is zero at $t = 0$.
- (2) Find the energy density u and the Poynting vector S in the gap as a function of s .
- (3) Determine the total energy U in the gap as a function of time.
- (4) Calculate the total power flowing into the gap by integrating the Poynting vector over the surface surrounding the gap.

4. Consider the reflection and transmission of light at oblique incidence as shown in the figure. The refractive indices in medium 1 and 2 are n_1 and n_2 , respectively.



- (1) Explain the "Plane of incidence" in the figure.

- (2) Write the relation between θ_I and θ_R .

- (3) Write the relation between θ_I and θ_T .

- (4) Explain the total internal reflection.

- (5) Express the critical angle θ_C over which the total internal reflection occurs using n_1 and n_2 .

- (6) Explain the evanescent wave.

Handwritten notes for question 4:
 $k_I \sin \theta_I = k_R \sin \theta_R = k_T \sin \theta_T$
 $\theta_I = \theta_R$
 $\frac{\sin \theta_T}{\sin \theta_I} = \frac{n_2}{n_1}$
 $\sin \theta_C = \frac{n_2}{n_1}$
 Evanescent wave $\rightarrow$ when total internal reflection occurs, there is a wave of no energy transmitted to the 2nd medium.

5. Electromagnetic plane waves of E and B that are incident into a conducting medium can be expressed as follows.

$$E(z, t) = E_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E) \hat{x}$$

$$B(z, t) = B_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E + \phi) \hat{y}$$

Answer the following questions.

(continued on the next page)

(1) Express the time-averaged energy density $\langle u \rangle$ of the plane wave using E_0 , B_0 , and other constants given in the equations.

(2) Use the following relation to express $\langle u \rangle$ without using B_0 .

$$\frac{B_0}{E_0} = \sqrt{\epsilon\mu} \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}$$

(3) Use the following relation to express $\langle u \rangle$ using k , ω , μ , κ , and E_0 .

$$k = \omega \sqrt{\frac{\epsilon\mu}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} + 1 \right]^{1/2}}$$

(4) Obtain the ratio of the time-averaged energy densities of the electric and magnetic waves and explain that the energy density of the latter is larger.

(5) Write the Poynting vector using the equations for $\mathbf{E}$ and $\mathbf{B}$ at the beginning.

(6) Express the time-averaged magnitude of the Poynting vector $\langle S \rangle$.

(7) Use the following relations and express the intensity of the electromagnetic wave without using B_0 and ϕ .

$$\frac{B_0}{E_0} = \frac{K}{\omega}$$

$$k = K \cos \phi$$

$$B_0 = \frac{k}{\omega} E_0$$